Toward Effective Lip Reading using Spatio-Temporal Encoding and Cluster-Guided Aggregation*

Luepol Pipanmekaporn[1], Kanabadee Srisomboon[2*], Wilaipon Lee[2], and Akara Prayote[1]

[1] Department of Computer and Information Science, King Mongkut's University of Technology North Bangkok, Bangkok, Thailand

[2] Department of Electrical and Computer Engineering, King Mongkut's University of Technology North Bangkok, Bangkok, Thailand

{luepol.p, kanabadee.s, wiliaporn.l, akara.p}@sci.kmutnb.ac.th

*Correspondence

Abstract

Lip reading aims to decode spoken words from visual lip movements, but it remains challenged by the high computational requirements and the difficulty of accurate discriminating lip motion patterns. Although recent spatio-temporal deep learning and transformer models have improved recognition performance, many rely on complex network architectures that limits practical deployment. To address these limitations, this study proposes an efficient and effective spatio-temporal lip reading framework that combines lightweight feature extraction with discriminative temporal aggregation. The proposed architecture combines a compact depthwise separable 3D Convolution with inspired residual blocks enhanced with Squeeze-and-Excitation (SE) modules to strengthen spatial representations with minimal overhead and a hierarchical Bidirectional GRU encoder to capture long-range temporal dependencies. To further improve robustness against visual ambiguity, NetVLAD-based cluster-guided aggregation is introduced to pool temporal descriptors according to learned visual word clusters, enabling the network to capture the complex lip-motion distributions. Evaluations on the Lip Reading in the Wild (LRW) benchmark demonstrate that the proposed model achieves a precision of 0.87, recall of 0.85, and F-score of 0.86. Additional evaluations on two specialized LRW subsets show strong robustness under challenging conditions, achieving F-scores of 0.74 on the HOMOPHONES dataset for visually confusable words

* Proceedings of The 2026 IFIP WG 8.4 International Symposium on E-Business Information Systems Evolution (EBISION 2026), Article No. 10, July 8–10, 2026, Seoul, South Korea.

© The copyright of this paper remains with the author(s).

and 0.84 on the INTER-S dataset for inter-speaker variability. Furthermore, the model requires only 21.05×10^9 FLOPs and 19.61×10^6 parameters, offering substantial computational savings compared to state-of-the-art baselines.

1 Introduction

Lip reading [1], commonly referred to as Visual Speech Recognition (VSR), involves automatically interpreting spoken content by analyzing the visual dynamics of lip movements. VSR serves a wide range of real-world applications, such as assisting individuals with hearing impairments [2], improving communication in noisy environments, and enabling automatic subtitle generation for multimedia content [3]. VSR also plays an important role in advancing human-computer interaction, strengthening surveillance capabilities, and supporting multimodal systems in situations where audio signals are unreliable or entirely absent [4].

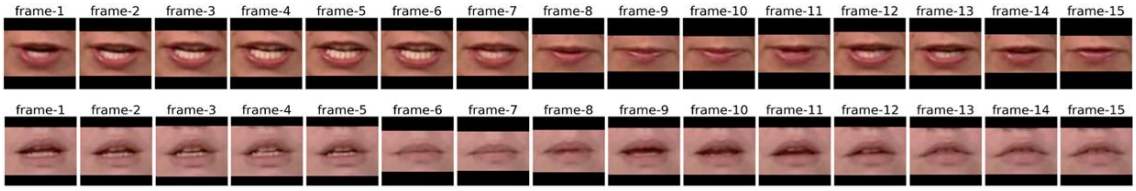
Despite recent advancements, lip reading still encounters some key challenges. One major difficulty is visual homophones ambiguity, where different words produce nearly identical lip movements. For example, words such as “PIN”, “PEN”, and “BIN” often appear visually similar, making them difficult for recognition models to distinguish. In addition, inter-speaker variability, including differences in lip shape, facial structure, and speaking speed, further increase the complexity of accurate recognition [6], [7]. Another significant challenge is the computational burden of modern deep learning-based architectures. Although these models achieve strong performance, they often rely on deep and complex networks that require substantial processing power and memory. This results in slow training and inference limiting deployment on real-time systems and resource-constrained devices such as assistive wearables [5]. Consequently, there is a growing need for lip reading approaches that are both efficient and accurate. These challenges are illustrated in Figure 1. .

In this study, we present an efficient lip-reading framework aimed at achieving a strong trade-off between recognition performance and computational cost. The proposed approach follows a hybrid spatio-temporal architecture, commencing with a lightweight 3D Convolutional Neural Network (3D CNN) to capture short-term spatio-temporal patterns from input video sequences. The extracted features are further enhanced through inverted residual blocks [8], which integrated with Squeeze-and-Excitation (SE) module to adaptively recalibrate channel-wise feature responses. This design strengthens spatial feature representation while maintaining a low computational footprint.

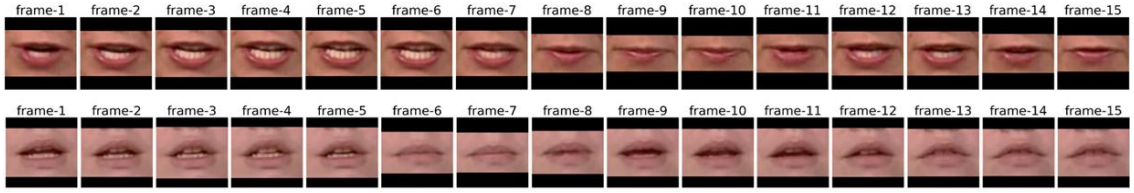
To model long-range temporal dependencies in lip-motion sequences, we employ a hierarchical Bidirectional Gated Recurrent Unit (Bi-GRU) encoder [9], enabling the network to learn contextual information in both forward and backward temporal directions. In addition, to improve temporal feature aggregation and address visual ambiguity, we integrate a learnable cluster-guided consensus module based on NetVLAD layer [10]. This module aggregates temporal descriptors according to their similarity to learned visual word clusters, allowing the model to capture complex lip movement distributions and improves discrimination between visually similar articulations. The proposed framework is evaluated on the Lip Reading in the Wild (LRW) dataset [11], as well as two specialized LRW subsets designed to assess robustness under challenging conditions: INTER-S for inter-speaker variability and HOMOPHONES for visually confusable words. Experimental results show that the proposed model achieves strong recognition performance while substantially reducing computational complexity compared with existing spatio-temporal and transformer-based lip-reading methods.

The main contributions of this work can be summarized as follows:

- We propose an efficient deep learning architecture that integrates a compact 3D CNN with inverted residual blocks with SE recalibration, and a hierarchical Bi-GRU encoder for effective and computationally efficient lip reading recognition.



(A) Video frame sequences annotated with the same word "BENEFIT" from two distinct speakers



(B) Video frame sequences annotated with the words "THING" (top row) and "THINK" (bottom row)

Fig. 1. Samples illustrating two key challenges: (A) Speaker variability and (B) Visually homophonic ambiguity

- We introduce a NetVLAD-based cluster guided aggregation to regenerate discriminative representations that improve robustness against lip-movement homophones.
- We evaluate the proposed model on the full LRW benchmark and two specialized LRW subsets, namely INTER-S and HOMOPHONES, which are designed to assess inter-speaker variability and visual lip-movement ambiguity. The results demonstrate strong recognition performance with reduced computational complexity, supporting the practical deployment of the proposed framework.

The subsequent sections are structured as follows: Section 2 examines relevant literature on approaches to lip reading. Section 3 provides a detailed description of the proposed methodology. Section 4 presents the experimental setup and reports the evaluation results. We conclude in Section 5 by summarizing our contributions and suggesting directions for future work.

2 Related work

2.1 Classical approaches

Early lip-reading methods primarily depended on handcrafted feature extraction combined with statistical sequence modeling. Traditional image processing techniques were employed to derive low-level visual cues [12], [13], including lip contours, shape descriptors, motion information, and appearance-based features such as the Discrete Cosine Transform (DCT) [14]and Local Binary Patterns (LBP) [15].These handcrafted descriptors were subsequently fed into statistical classifiers, such as Hidden Markov Models (HMMs) [16]and Dynamic Time Warping (DTW) [17], to perform sequence-level recognition. Although these approaches laid the groundwork for VSR, they suffered from limited generalization capability. Handcrafted features were highly sensitive to variations in speakers, lighting, and head pose, while statistical models were unable to fully capture the intricate temporal dynamics inherent in lip movements.

2.2 Deep learning-based approaches

The advent of deep learning has markedly advanced lip reading by enabling models to automatically learn both spatial and temporal representations directly from raw video data. Convolutional Neural Networks (CNNs) [18]have been extensively utilized to capture spatial characteristics of the lip region, while Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) units and Bidirectional Gated Recurrent Units (Bi-GRUs), have proven effective for modeling temporal dependencies across video frames [19]. To further boost recognition performance, recent work has investigated hybrid spatio-temporal architectures, such as 3D Convolutional Networks (Conv3D) [28] and Conv2D combined with Bi-GRU encoders [20], [21], [22], which jointly encode spatial and temporal cues. Moreover, transformer-based models [23], [24]have achieved strong results by capturing complex temporal dynamics and enhancing overall VSR capabilities.

Despite these advancements, deep learning–based VSR systems continue to encounter significant challenges related to computational complexity. Their reliance on deep and intricate architectures demands considerable processing power and memory, resulting in high training and inference costs. This computational burden restricts real-time deployment and increases hardware demands. Consequently, there is growing interest in developing efficient VSR architectures that can preserve competitive recognition accuracy while significantly reducing computational overhead, thereby facilitating broader and more practical use of VSR technologies. Nevertheless, visual homophones remain an unsolved issue that requires dedicated solutions.

3 Proposed Architecture

This section presents the proposed architecture, named Spatio-Temporal Deep Cluster Aggregation (ST-DCA), designed to achieve efficient lip reading while maintaining strong discriminative capability, particularly in the presence of visually similar lip movement patterns**.** As illustrated in Figure 2, the architecture comprises three key modules**: (**1**)** an efficient spatio-temporal feature extractor, **(**2**)** a

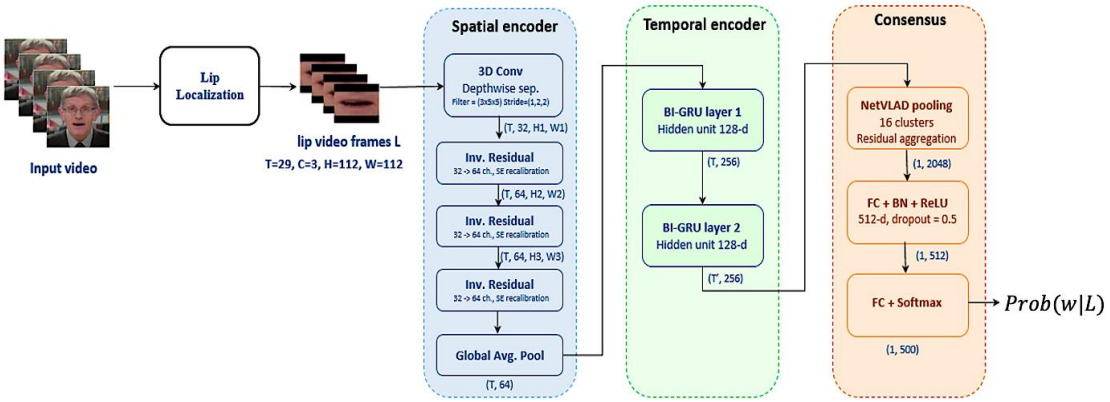


Fig. 2. The proposed ST-DCA framework for word-level lip reading. The input video is first processed by a lip localization module to extract a normalized mouth-region sequence. The spatial encoder then captures short-term spatio-temporal visual patterns using a depthwise separable 3D convolution followed by SE-enhanced inverted residual blocks. The resulting frame-level descriptors are passed to a hierarchical Bi-GRU temporal encoder to model long-range lip-motion dependencies. Finally, the NetVLAD-based consensus module aggregates the temporal features into a compact cluster-guided representation, which is fed to fully connected and softmax layers for final word classification.

hierarchical temporal encoder and **(**3**)** a consensus module that aggregates the temporal features for word recognition.

3.1 Lip Localization

The lip extraction pipeline operates on a frame-by-frame basis to focus specifically on the mouth region, reducing the risk of the model learning biases from irrelevant facial characteristics. In each frame, MediaPipe [25] is employed to detect 478 facial landmarks, from which 40 lip-related keypoints are selected. A bounding box is then computed using the extreme coordinates of these points, with an additional 10-pixel margin to ensure complete coverage of lip movements. The cropped region is subsequently centered and zero-padded to a fixed resolution of 112 × 112 pixels, maintaining a consistent input size for the network. The lip localization process is illustrated in Figure 3.

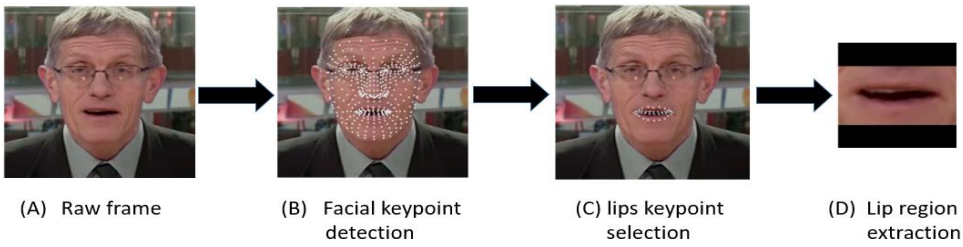


Fig 3. The lip localization process

3.2 Efficient Spatial Encoding

The spatial encoder is designed to extract discriminative representations from lip region sequences. Let the input sequence be denoted as a 4D tensor $X \in \mathbb{R}^{T\times C\times H\times W}$, where T is the number of timesteps, C is the channel count, and $H \times W$ represents the spatial resolution of each frame. While conventional VSR models typically rely on standard 2D or 3D convolutions, these operations are often computationally expensive and prone to filter redundancy, limiting both efficiency and model expressiveness. To mitigate these limitations, the proposed architecture employs a series of efficient convolutional blocks that reduce computational overhead without compromising performance.

The encoder begins with an efficient 3D convolutional block tailored for short-term spatiotemporal feature extraction. We replace the standard 3D convolution with a Depthwise Separable 3D Convolution [26], which factorizes the operation into two distinct stages: depthwise convolution and pointwise convolution. First, a depthwise step independently convolves each input channel using a 3D kernel with dimensions $K_t \times K_h \times K_w$ and a stride of (1, 2, 2). The operation is defined as:

$$D_c(t,h,w) = \sum_{i=0}^{K_t-1}\sum_{j=0}^{K_h-1}\sum_{k=0}^{K_w-1} W_{i,j,k}^{(c)} * X_c(t+i, h+j, w+k) \quad (1)$$

where $W^{(c)} \in \mathbb{R}^{K_t\times K_h\times K_w}$ denotes the 3D kernel specific to channel c and $D_c \in \mathbb{R}^{C_{in}\times K_t\times H\times W}$ is the corresponding output feature map. In this work, a kernel size of (3, 5, 5) is employed to effectively capture short-range lip movement dynamics. Subsequently, a pointwise convolution with a kernel size 1× 1× 1 fuses information across all channels from the depthwise outputs as defined below:

$$Y_c(t,h,w) = \sum_{c'=1}^{C_{in}} P_{c',c} * D_{c'}(t,h,w) \quad (2)$$

where $P_{c',c}$ is the weight mapping the c'-th the depthwise feature to the c-th output channel. Y_c denotes the final fused feature map. The resulting output is then passed through a batch normalization layer and a ReLU activation function to ensure training stability.

Following the initial 3D convolutional front-end, the encoder employs a stack of Inverted Residual blocks with Squeeze-and-Excitation (SE) modules to extract deep spatial representation. Inspired by MobileNetV3 [8], these blocks leverage an inverted residual design specifically tailored for efficient networks. This structure has been demonstrated to maximize parameter efficiency while preserving information flow better than standard residual blocks. As illustrated in Fig. 4, the process begins with a 1×1 pointwise convolution that expands the input feature map to a higher-dimensional intermediate space, facilitating the extraction of complex non-linear features. This is immediately followed by a depthwise convolution with a spatial kernel size of $k \times k$, which performs spatial filtering independently on each channel, drastically reducing computational cost compared to standard convolutions.To enhance the discriminative power of the network, a Squeeze-and-Excitation (SE) module is integrated after the depthwise layer to adaptively recalibrate channel-wise importance. This mechanism compresses global spatial information using Global Average Pooling (GAP) and recalibrates channel responses using a compact fully connected network, effectively emphasizing relevant lip motion patterns while suppressing background noise. Finally, a second 1×1 pointwise convolution projects the calibrated features back to a lower target dimensionality. A residual skip connection is also employed to add the original input directly to the output, ensuring stable gradient propagation and preserving information flow throughout the deep network.

3.3 Hierarchical Temporal Encoding

Following the spatial encoding stage, the network employs a temporal modeling module to capture long-range temporal dynamics of lip movements across the video sequence. To achieve this, we utilize multiple layers of Bidirectional Gated Recurrent Units (BiGRUs) [9] to extract hierarchical temporal representations. Unlike a standard GRU, a BiGRU consists of two recurrent units operating in opposite temporal directions: a forward GRU that processes the sequence from $t = 1$ to T, and a backward GRU that proceeds from $t = T$ to 1. For the forward direction, the update gate, reset gate, candidate state, and hidden state are computed as:

$$\begin{aligned}
\overrightarrow{z_t} &= \sigma\left(W_z^f x_t + U_z^f \overrightarrow{h_{t-1}} + b_z^f\right) \quad (3)\\
\overrightarrow{r_t} &= \sigma\left(W_r^f x_t + U_r^f \overrightarrow{h_{t-1}} + b_r^f\right)\\
\overrightarrow{\widetilde{h_t}} &= \tanh\left(W_h^f x_t + U_r^f(\overrightarrow{r} \odot \overrightarrow{h_{t-1}}) + b_h^f\right)\\
\overrightarrow{h_t} &= (1 - \overrightarrow{z_t}) \odot \overrightarrow{h_{t-1}} + \overrightarrow{z_t} \odot \overrightarrow{\widetilde{h_t}}
\end{aligned}$$

Similarly, the backward GRU follows the same formulation but replaces $\overrightarrow{h_{t-1}}$ with $\overleftarrow{h_{t-1}}$, as it operates in the reverse temporal direction. The final bidirectional representation at time t is obtained by concatenating the two directional states $h_t = \left[\overrightarrow{h_t}; \overleftarrow{h_t}\right]$, which provides a unified feature embedding incorporating context information from both frame directions. To expand the temporal

receptive field without excessive parameter growth, a second-level BiGRU processes a temporally down-sampled sequence:

$$\widetilde{h_t'} = BiGRU_2\left(h_{st'}\right) \quad s > 1 \qquad (4)$$

where $h_{st'}$ represents the outputs of the first-level BiGRU, subsampled every s time steps to shorten the sequence. This operation applies a higher-level BiGRU to a compressed temporal input, allowing the model to capture longer-range dependencies while keeping the parameter and computational costs low.

3.4 Cluster-Guided Aggregation

To leverage the temporal features for final prediction, a common approach is to average the features across all time steps, producing a single fixed-length consensus vector. While this aggregation strategy offers computational simplicity, it relies on the unrealistic assumption that all frames contribute equally to the final decision. As illustrated in Fig. 1(B), only timesteps 2 and 15 carry meaningful information for the word 'THINK,' even though the sample contains 29 frames. Such uniform weighting can diminish important information and may introduce noise from irrelevant frames. To obtain a compact and discriminative utterance-level representation, we propose to incorporate a Deep-Cluster consensus module based on a learnable NetVLAD aggregation mechanism [10], which adaptively emphasizes informative features while suppressing non-discriminative ones. Given the sequence of temporal features $\tilde{h} = \{\tilde{h}_1, \tilde{h}_2, \dots, \tilde{h}_N\}$, $\tilde{h}_i \in \mathbb{R}^D$ produced by the hierarchical BiGRU encoder and the module learns a set of K clusters $= \{c_1, c_2, \dots, c_K\}$ $c_i \in \mathbb{R}^D$, which act as prototypical lip-movement patterns. Each feature $\tilde{h}_i$, is softly assigned to every cluster using a linear scoring function:

$$a_{ik} = \frac{\exp\left(w_k^T \tilde{h}_i + b_k\right)}{\sum_{j=1}^{K} \exp\left(w_j^T \tilde{h}_i + b_j\right)} \qquad (5)$$

where w_k is a trainable parameter vector associated with cluster k and b_k denotes its corresponding bias term. Using these assignments, NetVLAD aggregates residuals relative to each cluster center:

$$V_k = \sum_{i=1}^{N} a_{ik}\left(\tilde{h}_i - c_k\right) \qquad (6)$$

and forms the final consensus representation by concatenating all cluster residuals:

$$V = [V_1; V_2; \dots; V_K] \qquad (7)$$

The resulting descriptor is passed through a fully connected layer with 512 units and dropout regularization, followed by a softmax classifier that produces the final probability distribution over target word classes.

4 Experiments

4.1 Datasets

The Lip Reading in the Wild (LRW) dataset [11] is used as the primary large-scale benchmark for word-level lip reading. LRW contains 538,766 short video clips extracted from BBC broadcasts, with each clip lasting approximately 1.16 seconds. The dataset includes more than 1,000 speakers and covers 500 target word classes. In this experiment, we follow the standard LRW split, which includes 488,766 training samples, 25,000 validation samples, and 25,000 test samples.

To evaluate two major challenges in lip reading, including visual homophone ambiguity and inter-speaker variability, speaker-level information is required for each video clip. Therefore, in this study, we utilize the speaker metadata provided by [27] to construct controlled evaluation subsets from LRW. These subsets are designed to isolate specific sources of difficulty and provide a more rigorous assessment of model robustness under challenging visual speech recognition conditions.

To assess the model's ability to distinguish visually confusable words, we introduce the HOMOPHONES evaluation benchmark, which is constructed from the broader LRW dataset. This subset focuses on word pairs that exhibit highly similar lip movements rather than phonetic homophones in the conventional linguistic sense. The word pairs were manually selected based on three criteria: (1) high visual similarity in mouth articulation, particularly during the initial or final lip movements; (2) practical ambiguity commonly observed in word-level lip-reading tasks, including singular–plural forms, morphological variants, and phoneme-close word pairs; and (3) sufficient sample availability in the LRW test partition to support reliable evaluation. As shown in Table 1, the HOMOPHONES subset consists of 60 expert-selected words forming 30 highly confusable word pairs, such as "BETTER" / "MATTER," "POLITICAL" / "POLITICS," and "THEIR" / "THERE." By parsing the LRW annotation files, all instances corresponding to these 60 target words were identified from the standard LRW test partition, resulting in a total of 3,791 test samples.

A – C	D – N	P - S	T – W
BECAUSE / BEFORE	DECISION / DETAILS	PAYING / SPENDING	TAKING / TALKING
BETTER / MATTER	DIFFERENT / DIFFICULT	PLACES / PLACE	THEIR / THERE
BILLION / MILLION	DOING / GOING	POLITICAL / POLITICS	THINGS / THESE
CHANGE / CHANCES	FAMILIES / FAMILY	POWERS / POWER	THOUGHT / THROUGH
CHILDREN / CHILD	FIGHTING / FIGHT	PROBLEM / PROBLEMS	TODAY / TOGETHER
COMPANIES / COMPANY	FIRST / FURTHER	SPEAKING / SPENDING	WHETHER / WEATHER
CONTROL / COUNCIL	FRANCE / FRENCH		WEEKS / WEEKEND
COUNTRIES / COUNTRY	HEART / HEARD		WORDS / WORLD

Table 1: Thirty manually selected visually confusable word pairs from the LRW dataset, grouped alphabetically.

Although this benchmark does not cover every possible form of visual homophonic ambiguity, it provides a controlled and challenging subset for evaluating fine-grained discrimination between visually similar lip movements. To prevent data leakage, the Speaker_ID metadata is used to enforce strict mutual exclusivity between the training/validation splits and the test split for the selected word pairs. Models are trained on the standard LRW training and validation sets after excluding speakers that overlap with the HOMOPHONES test subset, yielding 474,352 training samples and 21,349 validation samples, respectively.

To evaluate speaker-independent robustness across the full 500-word LRW vocabulary, we introduce an additional evaluation subset, denoted as INTER-S. This subset is designed to examine whether a model can generalize to speakers who are completely unseen during training and validation. Using the Speaker_ID metadata provided by [27]. we first analyzed the word-class coverage of each speaker. Since no individual speaker in the available metadata contained samples for all 500 target word classes, we selected 10 speakers with broad word-class coverage to construct a challenging unseen-

speaker evaluation subset. The selected speaker IDs are 237, 4464, 6850, 7184, 8334, 9438, 9871, 10506, 11146, and 16585. All clips from these selected speakers were assigned exclusively to the INTER-S test subset. To ensure a strict unseen-speaker evaluation and prevent speaker-level data leakage, every training and validation instance associated with these 10 speakers was removed from the standard LRW partitions. As a result, the INTER-S test set contains 5,787 samples, while the pruned training and validation sets contain 450,955 and 23,632 samples, respectively.

4.2 Data Preprocessing

For preprocessing, MediaPipe [25] was employed to detect 478 facial landmarks in each frame. Among these, 40 keypoints corresponding to the lip region were selected to define a bounding box, which expanded with a 10-pixel margin to fully capture lip movements. The extracted region was then centered and zero-padded to a fixed resolution of 112×112 pixels, reducing potential bias introduced by speaker-specific facial characteristics. No data augmentation was applied during training on either the INTER-S or HOMOPHONES datasets to ensure a fair and controlled comparison with all baseline architectures.

4.3 Baselines

We evaluated the proposed ST-DCA model against widely used spatio-temporal lip reading architectures:

- **Conv3D** [28]: An early spatio-temporal model that employs stacked 3D convolutional layers to jointly encode spatial and temporal information from raw video inputs. Temporal average pooling is used for sequence aggregation, followed by a softmax classifier for word-level prediction.
- **ResNet18 + Bi-GRU** [23]: A hybrid architecture in which ResNet-18 extracts frame-level spatial features from mouth-region images. These features are then processed by bidirectional GRUs to capture temporal dependencies, with word classification performed via softmax.
- **Conv3D + Bi-GRU** [21]: A two-stage hybrid model combining 3D convolutions for short-term temporal modeling with a two-layer Bi-GRU module for long-range sequence encoding.
- **ViT + Temporal Transformer** [24]: A transformer-based architecture where each frame is encoded using a Vision Transformer (ViT), and temporal modeling is performed using a multi-head self-attention transformer applied across the sequence.
- **Conv3D -2D + Bi-GRU** [22]: A hybrid approach that integrates 3D convolutions with 2D residual blocks for spatial feature extraction, followed by stacked Bi-GRUs for sequential modeling.

4.4 Implementation Details

The spatial encoder begins with a Depthwise Separable 3D convolution with 32 output channels, a 3×5×5 kernel, and a stride 1×2×2. This is followed by three inverted residual blocks with Squeeze-and-Excitation (SE) modules, progressively increasing channel capacity from 32 to 64. Global average pooling is applied to each frame to produce a temporal feature sequence of size 27×64. The sequence is then processed by a hierarchical two-layer BiGRU with a hidden dimension of 128 in each direction. The first BiGRU layer produces a 256-dimensional temporal feature at each time step. The second BiGRU layer is applied to a temporally down-sampled sequence using a subsampling rate of ($s = 2$).

The final BiGRU output remains 256-dimensional at each time step. The NetVLAD module with K = 16 clusters aggregates these features into a 4,096-dimensional global descriptor. This descriptor is then passed through a fully connected layer with 512 units, followed by BatchNorm, ReLU activation, and dropout with p = 0.5, before the final softmax classifier.

All models were optimized using AdamW with a weight decay of 1×10^{-4}, $\beta_1 = 0.9$, and $\beta_2 = 0.98$. The initial learning rate was set to 1×10^{-4}, with linear warm-up for 8 epochs on LRW and 12 epochs on HOMOPHONES, followed by cosine annealing decay. Training was performed with a batch size of 16 and early stopping based on validation F1-score with a patience of 30 epochs. All experiments were conducted on a single NVIDIA T4 GPU, with training requiring approximately 130–330 hours for LRW and 20–50 hours for HOMOPHONES.

4.5 Results and Discussion

Quantitative comparisons on the full LRW benchmark are presented in Table 2. Overall, the proposed ST-DCA model achieves the highest recognition performance among all evaluated methods, with a precision of 0.87, recall of 0.85, and F-score of 0.86. Compared with the transformer-based ViT+Temporal Transformer baseline, ST-DCA improves the F-score from 0.85 to 0.86 while requiring substantially fewer computational resources. Specifically, ST-DCA uses only 21.05×10^9 FLOPs and 19.61×10^6 parameters, whereas ViT+Temporal Transformer requires 36.89×10^9 FLOPs and 39.7×10^6 parameters. This corresponds to approximately 42.9% fewer FLOPs and 50.6% fewer parameters, demonstrating that ST-DCA provides a more favorable balance between recognition accuracy and model efficiency.

Further evaluations on the HOMOPHONES and INTER-S datasets are summarized in Table 3. On the HOMOPHONES dataset, which contains visually confusable word classes, ST-DCA achieves the best performance with a precision of 0.75, recall of 0.73, and F-score of 0.74. This result outperforms the strongest baseline, ViT+Temporal Transformer, which obtains an F-score of 0.71. The improvement indicates that ST-DCA is more effective in distinguishing subtle lip-motion differences among words with highly similar visual patterns. Compared with Conv3D+Bi-GRU and Conv3D-2D+Bi-GRU, ST-DCA also shows clear gains, suggesting that the proposed spatio-temporal discriminative feature aggregation contributes to more robust classification under visually ambiguous conditions.

On the INTER-S dataset, ST-DCA again achieves the best overall performance, reaching a precision of 0.85, recall of 0.84, and F-score of 0.84. Although ViT+Temporal Transformer performs competitively with an F-score of 0.83, ST-DCA still provides a slight improvement while maintaining a considerably lighter computational profile, as shown in Table 3. This confirms that the proposed model generalizes well across inter-speaker variations and diverse word classes without relying on a large transformer-based architecture.

The improved discriminative capability of ST-DCA is further supported by the ROC curves shown in Figure 5. On the HOMOPHONES dataset, ST-DCA produces a steeper ROC curve than the

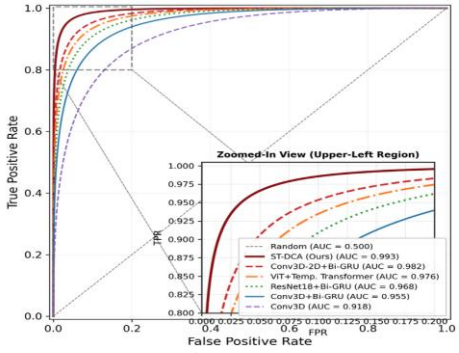
Method	Full LRW			FLOPS ($1x10^9$)	#Param ($1x10^6$)
	Precision	Recall	F-score		
Conv3D	0.71	0.68	0.69	48.53	17.1
ResNet18+Bi-GRU	0.83	0.81	0.82	16.8	33.54
Conv3D+Bi-GRU	0.79	0.75	0.77	22.75	22.0
ViT +Temp. Transformer	0.86	0.84	0.85	36.89	39.7
Conv3D-2D+Bi-GRU	0.85	0.83	0.84	26.4	28.4
ST-DCA	**0.87**	**0.85**	**0.86**	**21.05**	**19.61**

TABLE 2. Performance comparison of ST-DCA with existing methods on the full LRW benchmark.

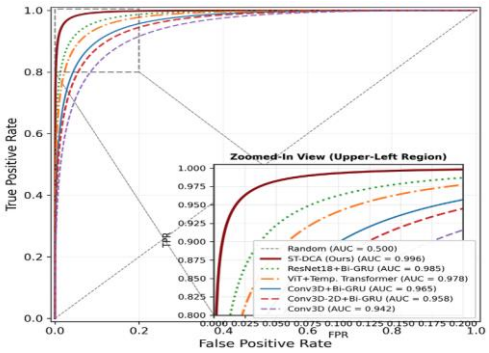
baseline models, indicating higher true positive rates at lower false positive rates. A similar trend is observed on the INTER-S dataset, where ST-DCA maintains consistently strong detection performance across different decision thresholds. The proposed model achieves AUC values of 0.993 and 0.996 on the HOMOPHONES and INTER-S datasets, respectively, outperforming the compared methods. These results suggest that ST-DCA learns more reliable and well-separated decision boundaries, particularly in challenging scenarios involving visually similar words and inter-speaker variability.

TABLE 3. Performance comparison of ST-DCA with existing methods on the HOMOPHONES and INTER-S datasets.

Method	HOMOPHONES			INTER-S		
	Precision	Recall	F-score	Precision	Recall	F-score
Conv3D	0.53	0.50	0.51	0.58	0.57	0.57
ResNet18+Bi-GRU	0.51	0.50	0.50	0.73	0.72	0.72
Conv3D+Bi-GRU	0.63	0.60	0.61	0.66	0.62	0.64
ViT+Temp. Transformer	0.72	0.72	0.71	0.84	0.83	0.83
Conv3D-2D+Bi-GRU	0.69	0.67	0.68	0.81	0.79	0.80
ST-DCA	**0.75**	**0.73**	**0.74**	**0.85**	**0.84**	**0.84**



(A) ROC Curve for HOMOPHONES classes



(B) ROC Curve for Inter-S classes

Fig. 5. ROC curves comparing the performance of different network architectures for lip reading on (A) the HOMOPHONES and (B) the INTER-S datasets. A zoomed-in inset of the upper-left region is provided to improve readability.

Beyond the overall quantitative results, we further examine performance at a finer level. Figure 6 illustrates the pairwise error rate (PER) comparison between the proposed ST-DCA and the transformer baseline on the most visually confusable word pairs in the HOMOPHONES dataset. Lower PER indicates better discrimination performance. As shown in this figure, ST-DCA consistently achieves lower error rates than the transformer model across most evaluated word pairs. The improvement is particularly evident for plural–singular and morphologically similar pairs such as FAMILIES/FAMILY, PROBLEM/PROBLEMS, and COMPANIES/COMPANY, where ST-DCA (green bar) substantially reduces misclassification rates. In contrast, the transformer baseline (gray bar) exhibits higher variability and consistently larger error rates across these challenging categories.

Moreover, the mean PER of ST-DCA (24.2%) is significantly lower than that of the transformer baseline (38.0%). These findings demonstrate that the proposed cluster-guided spatio-temporal architecture is more effective at capturing fine-grained temporal cues, leading to improved discrimination of visually similar word pairs. Figure 7 compares the per-class accuracy distributions of ST-DCA and the ViT-Transformer baseline across the 500 word classes on the INTER-S benchmark. Although both models achieve high accuracy for most the classes, ST-DCA is more concentrated in the 85-95% accuracy range and shows a clear shift toward higher performance. In contrast, the transformer baseline has a wider distribution with more classes falling below 60%, indicating less consistent recognition across difficult categories.

These results confirm that ST-DCA not only improves overall recognition accuracy, but also provides more consistent and reliable performance across diverse word classes, reducing the number of poorly recognized categories compared to the transformer-based approach.

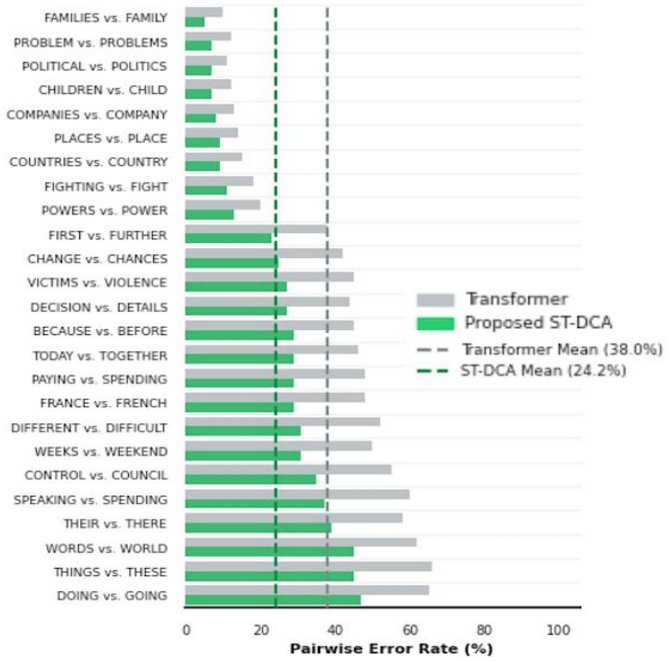


Fig. 6 . Comparison of pairwise error rates (PER) between the Temporal Transformer (gray) and the ST-DCA model (green) on homophone pairs with PER below 60%

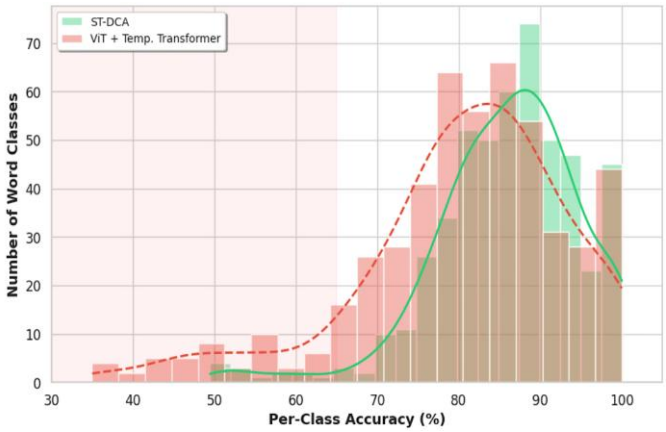


Fig. 7. Per-class accuracy distribution of the ST-DCA and the transformer baseline across the 500-word INTER-S dataset

Table 4 presents the ablation study on the NetVLAD cluster count (K) with results reported as the mean ± standard deviation (SD) across three independent training runs with random seeds (42, 132, 2024). Increasing K from 4 to 16 consistently improves performance on HOMOPHONES, with K=16 achieving the best HOMOPHONES result of 0.740±0.002; on INTER-S, K=32 achieves the highest mean F1-score of 0.841±0.006, while K=16 reaches 0.840±0.011.The low standard deviations across all configurations demonstrate that the ST-DCA architecture is highly robust to weight initialization; the slight variance increase at K=32 is observed on HOMOPHONES, whereas the INTER-S standard

Dataset	Number of K Clusters (Mean F1 ± Standard Deviation)			
	4	8	16	32
HOMOPHONES	0.650 ± 0.004	0.701± 0.003	0.740± 0.002	0.731 ± 0.004
INTER-S	0.794 ± 0.002	0.810 ± 0.004	0.840 ± 0.011	0.841 ±0.006

Table 4 –F1-score performance of ST-DCA with varying NetVLAD cluster sizes (K)

deviation decreases from 0.011 at K=16 to 0.006 at K=32.

Overall, the result confirms that the choice of cluster count plays a critical role in shaping the effectiveness of temporal aggregation within the ST-DCA architecture. Specifically, smaller cluster sizes fail to encode the complex distributions required for homophone disambiguation, while excessively large clusters introduce redundant representations. The K=16 configuration provides a strong trade-off for HOMOPHONES while maintaining nearly the same INTER-S mean F1-score as K=32, helping isolate fine-grained articulation differences without sacrificing computational efficiency.

5 Conclusion

This study introduced the ST-DCA framework to tackle two persistent bottlenecks in lip reading: the high computational cost of modern architectures and the difficulty of disambiguating visual homophones. By integrating a lightweight spatial encoder—utilizing depthwise-separable convolutions and SE-enhanced inverted residual blocks—with a hierarchical Bi-GRU for temporal modeling, the framework efficiently captures long-range dependencies. The inclusion of a NetVLAD-based cluster-guided consensus module further empowers the model to map complex lip movement distributions,

significantly boosting discrimination against visually similar words. Consequently, ST-DCA extracts highly discriminative articulatory features with a fraction of the computational overhead required by conventional CNN-RNN and transformer baselines. Evaluations on the unseen-speaker INTER-S subset of LRW and the newly curated HOMOPHONES dataset validate the model's state-of-the-art recognition capabilities.

Several directions remain open for future investigation. First, the proposed framework is designed for word-level visual speech recognition. Extending ST-DCA to sentence-level recognition represents an important direction for enhancing its practical applicability in real-world communication scenarios, where continuous and context-dependent speech patterns are more common. Second, the current NetVLAD-based clustering module adopts a fixed number of clusters determined empirically through experiments. Future work will investigate adaptive or hierarchical clustering strategies that can dynamically adjust the number of visual prototypes according to input complexity, thereby improving discriminative capability across a wider range of lip-motion patterns. Finally, although this study focuses on visual-only recognition, integrating ST-DCA into a multimodal audio-visual framework could further improve robustness, particularly under acoustically noisy or visually ambiguous conditions.

Acknowledgement

This research was supported by the National Science, Research and Innovation Fund (NSRF) and King Mongkut’s University of Technology North Bangkok, under Project No. KMUTNB-FF-69-B-15.

References

[1] Nemani, P., Krishna, G. S., Supriya, K., & Kumar, S. (2023). Speaker independent VSR: A systematic review and futuristic applications. Image and Vision Computing, 138, Article 104787.

[2] Ivanko, D., Ryumin, D., & Karpov, A. (2019). Automatic lip-reading of hearing impaired people. The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 42, 97–91.

[3] Kavitha, R., Dinesh Babu, P. K., & Yashwanth, J. A. (2024). Realistic lip synchronization using deep learning techniques for speech generation. In Proceedings of the 2024 International BIT Conference (BITCON) (pp. 1–6). IEEE.

[4] Jeevakumari, S. A. A., & Dey, K. (2024). LipSyncNet: A novel deep learning approach for visual speech recognition in audio-challenged situations. IEEE Access, 12, 1–14.

[5] Yang, C., Wang, S., Zhang, X., & Zhu, Y. (2020). Speaker-independent lipreading with limited data. In Proceedings of the 2020 IEEE International Conference on Image Processing (ICIP) (pp. 2181–2185). IEEE.

[6] Chotikkakamthorn, K., Ritthipravat, P., Kusakunniran, W., Tuakta, P., & Benjapornlert, P. (2022). A lightweight deep learning approach to mouth segmentation in color images. Applied Computing and Informatics.

[7] Wen, J., & Lu, Y. (2019). Automatic lip reading system based on a fusion lightweight neural network with Raspberry Pi. Applied Sciences, 9(24), Article 5432.

[8] Howard, A., Sandler, M., Chu, G., Chen, L.-C., Chen, B., Tan, M., Wang, W., Zhu, Y., Pang, R., Vasudevan, V., Le, Q. V., & Adam, H. (2019). Searching for MobileNetV3. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) (pp. 1314–1324). IEEE.

[9] Cho, K., Van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using RNN encoder–decoder for statistical machine translation. In Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP) (pp. 1724–1734). ACL.

[10] Arandjelović, R., Gronat, P., Torii, A., Pajdla, T., & Sivic, J. (2016). NetVLAD: CNN architecture for weakly supervised place recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (pp. 5297–5307). IEEE.

[11] Chung, J. S., & Zisserman, A. (2017). Lip reading in the wild. In Computer Vision – ACCV 2016: 13th Asian Conference on Computer Vision, Revised Selected Papers, Part II (pp. 87–103). Springer.

[12] Dalka, P., Bratoszewski, P., & Czyzewski, A. (2014). Visual lip contour detection for the purpose of speech recognition. In Proceedings of the 2014 International Conference on Signals and Electronic Systems (ICSES) (pp. 1–4). IEEE.

[13] Tian, Y., Kanade, T., & Cohn, J. (2000). Robust lip tracking by combining shape, color and motion. In Proceedings of the 4th Asian Conference on Computer Vision (pp. 1040–1045).

[14] Jain, A., & Rathna, G. N. (2017). Visual speech recognition for isolated digits using discrete cosine transform and local binary pattern features. In Proceedings of the 2017 IEEE Global Conference on Signal and Information Processing (GlobalSIP) (pp. 368–372). IEEE.

[15] Zhao, G., Barnard, M., & Pietikainen, M. (2009). Lipreading with local spatiotemporal descriptors. IEEE Transactions on Multimedia, 11(7), 1254–1265

[16] Sharma, U., Maheshkar, S., Mishra, A. N., & Kaushik, R. (2019). Visual speech recognition using optical flow and hidden Markov model. Wireless Personal Communications, 106, 2129–2147.

[17] Lee, K., Keum, M., Han, D. K., & Ko, H. (2015). Visual speech recognition using weighted dynamic time warping. IEICE Transactions on Information and Systems, 98(7), 1430–1433.

[18] Chan, Z.-M., Lau, C. Y., & Thang, K.-F. (2020). Visual speech recognition of lips images using convolutional neural network in VGG-M model. Journal of Information Hiding and Multimedia Signal Processing, 11(3), 116–125.

[19] Chen, X., Wang, C., Li, Y., Hu, C., Wang, Q., & Cai, D. (2023). Research on bidirectional LSTM recurrent neural network in speech recognition. In Proceedings of the 2023 IEEE 6th International Conference on Pattern Recognition and Artificial Intelligence (PRAI) (pp. 878–883). IEEE.

[20] Fernandez-Lopez, A., & Sukno, F. M. (2018). Survey on automatic lip-reading in the era of deep learning. Image and Vision Computing, 78, 53–72.

[21] Assael, Y. M., Shillingford, B., Whiteson, S., & De Freitas, N. (2016). LipNet: Sentence-level lipreading. arXiv. https://arxiv.org/abs/1611.01599

[22] Feng, D., Yang, S., Shan, S., & Chen, X. (2020). Learn an effective lip reading model without pains. arXiv. https://arxiv.org/abs/2011.07557

[23] Ma, P., Petridis, S., & Pantic, M. (2021). End-to-end audio-visual speech recognition with conformers. In Proceedings of the 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 7613–7617). IEEE.

[24] Xue, F., Li, Y., Liu, D., Xie, Y., Wu, L., & Hong, R. (2023). LipFormer: Learning to lipread unseen speakers based on visual-landmark transformers. IEEE Transactions on Circuits and Systems for Video Technology, 33(9), 4507–4517.

[25] Lugaresi, C., Tang, J., Nash, H., McClanahan, C., Uboweja, E., Hays, M., Zhang, F., Chang, C.-L., Yong, M. G., Lee, J., Chang, W.-T., Hua, W., Georg, M., & Grundmann,

M. (2019). MediaPipe: A framework for building perception pipelines. arXiv. https://arxiv.org/abs/1906.08172

[26] Srivastava, H., & Sarawadekar, K. (2020). A depthwise separable convolution architecture for CNN accelerator. In Proceedings of the 2020 IEEE Applied Signal Processing Conference (ASPCON) (pp. 1–5). IEEE.

[27] Kim, M., Kim, H., & Ro, Y. M. (2022). Speaker-adaptive lip reading with user-dependent padding. In European Conference on Computer Vision (pp. 576–593). Springer.

[28] Tran, D., Bourdev, L., Fergus, R., Torresani, L., & Paluri, M. (2015). Learning spatiotemporal features with 3D convolutional networks. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)* (pp. 4489–4497). IEEE.